



Algorithm and Hardness Results on Liar's Dominating Set and k -tuple Dominating Set

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Abstract. Given a graph $G = (V, E)$, the dominating set problem asks for a minimum subset of vertices $D \subseteq V$ such that every vertex $u \in V \setminus D$ is adjacent to at least one vertex $v \in D$. That is, the set D satisfies the condition that $|N[v] \cap D| \geq 1$ for each $v \in V$, where $N[v]$ is the closed neighborhood of v . In this paper, we study two variants of the classical dominating set problem: k -tuple dominating set (k -DS) problem and Liar's dominating set (LDS) problem, and obtain several algorithmic and hardness results. On the algorithmic side, we present a constant factor $(\frac{11}{2})$ -approximation algorithm for the Liar's dominating set problem on unit disk graphs. Then, we design a polynomial time approximation scheme (PTAS) for the k -tuple dominating set problem on unit disk graphs. On the hardness side, we show a $\Omega(n^2)$ bits lower bound for the space complexity of any (randomized) streaming algorithm for Liar's dominating set problem as well as for the k -tuple dominating set problem. Furthermore, we prove that the Liar's dominating set problem on bipartite graphs is W[2]-hard.

1 Introduction

The dominating set problem is regarded as one of the fundamental problems in theoretical computer science which finds its applications in various fields of science and engineering [3, 8]. A *dominating set* of a graph $G = (V, E)$ is a subset D of V such that every vertex in $V \setminus D$ is adjacent to at least one vertex in D . The *domination number*, denoted as $\gamma(G)$, is the minimum cardinality of a dominating set of G . Garey and Johnson [6] showed that deciding whether a given graph has domination number at most some given integer k is NP-complete. For a vertex $v \in V$, the open neighborhood of the vertex v denoted as $N_G(v)$ is defined as $N_G(v) = \{u \mid (u, v) \in E\}$ and the closed neighborhood of the vertex v denoted as $N_G[v]$ is defined as $N_G[v] = N_G(v) \cup \{v\}$.

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***k*-tuple Dominating Set (*k*-DS):** Fink and Jacobson [5] generalized the concept of dominating sets as follows.

***k*-tuple Dominating Set (*k*-DS) Problem**

Input: A graph $G = (V, E)$ and a non-negative integer k .

Goal: Choose a minimum cardinality subset of vertices $D \subseteq V$ such that, for every vertex $v \in V$, $|N_G[v] \cap D| \geq k$.

The *k*-tuple domination number $\gamma_k(G)$ is the minimum cardinality of a *k*-DS of G . A good survey on *k*-DS can be found in [4, 12]. Note that, 1-tuple dominating set is the usual dominating set, and 2-tuple and 3-tuple dominating set are known as *double dominating set* [7] and *triple dominating set* [15], respectively. Further note that, for a graph $G = (V, E)$, $\gamma_k(G) = \infty$, if there exists no *k*-DS of G . Klasing et al. [10] studied the *k*-DS problem from hardness and approximation point of view. They gave a $(\log |V| + 1)$ -approximation algorithm for the *k*-tuple domination problem in general graphs, and showed that it cannot be approximated within a ratio of $(1 - \epsilon) \log |V|$, for any $\epsilon > 0$.

Liar's Dominating Set (LDS): Slater [17] introduced a variant of the dominating set problem called Liar's dominating set problem. Given a graph $G = (V, E)$, in this problem the objective is to choose minimum number of vertices $D \subseteq V$ such that each vertex $v \in V$ is double dominated and for every two vertices $u, v \in V$ there are at least three vertices in D from the union of their neighborhood set. The LDS problem is an important theoretical model for the following real-world problem. Consider a large computer network where a virus (generated elsewhere in the Internet) can attack any of the processors in the network. The network can be viewed as an unweighted graph. For each node $v \in V$, an anti-virus can: (1) detect the virus at v as well as in its closed neighborhood $N[v]$, and (2) find and report the vertex $u \in N[v]$ at which the virus is located. Notice that, one can make network G virus free by deploying the anti-virus at the vertices $v \in D$, where D is the minimum size *dominating set*. However, in certain situations the anti-viruses may fail. Hence, to make the system virus free it is likely to double-guard the nodes of the network, which is indeed the 2-tuple DS. However, despite of the double-guarding, the anti-viruses may fail to cure the system properly due to some software error or corrupted circumstances. Therefore, for every pair of nodes, it is be important to introduce a guard that sees both of them. This leads us to the Liar's dominating set problem. We define the problem formally below.

Liar's Dominating Set (LDS) Problem

Input: A graph $G = (V, E)$ and a non-negative integer k .

Goal: Choose a subset of vertices $L \subseteq V$ of minimum cardinality such that for every vertex $v \in V$, $|N_G[v] \cap L| \geq 2$, and for every pair of vertices $u, v \in V$ of distinct vertices $|(N_G[u] \cup N_G[v]) \cap L| \geq 3$.

1.1 Our Results

In this paper, we obtain several algorithmic and hardness results for LDS and k -DS problems on various graph families. On the algorithmic side in Sect. 2, we present a constant factor $(\frac{11}{2})$ -approximation algorithm for the Liar's dominating set (LDS) problem on unit disk graphs. Then, we design a polynomial time approximation scheme (PTAS) for the k -tuple dominating set (k -DS) problem on unit disk graphs. On the hardness side in Sect. 3, we show a $\Omega(n^2)$ bits lower bound for the space complexity of any (randomized) streaming algorithm for Liar's dominating set problem as well as for the k -tuple dominating set problem. Furthermore, we prove that the Liar's dominating set problem on bipartite graphs is $W[2]$ -hard.

2 Algorithmic Results

2.1 Approximation Algorithm for LDS on Unit Disk Graphs

Unit disk graphs are widely used to model of wireless sensor networks. A unit disk graph (UDG) is an intersection graph of a family of unit radius disks in the plane. Formally, given a collection $\mathcal{C} = \{C_1, C_2, \dots, C_n\}$ of n unit disks in the plane, a UDG is defined as a graph $G = (V, E)$, where each vertex $u \in V$ corresponds to a disk $C_u \in \mathcal{C}$ and there is an edge $(u, v) \in E$ between two vertices u and v if and only if their corresponding disks C_u and C_v contain v and u , respectively. Here, we study the LDS problem on UDG.

Liar's Dominating Set on UDG (*LDS-UDG*) Problem

Input: A unit disk graph $G = (\mathcal{P}, E)$, where $\mathcal{P}$ is a set of n disk centers.

Output: A minimum size subset $D \subseteq \mathcal{P}$ such that for each point $p_i \in \mathcal{P}$, $|N[p_i] \cap D| \geq 2$, and for each pair of points $p_i, p_j \in \mathcal{P}$, $|(N[p_i] \cup N[p_j]) \cap D| \geq 3$.

Jallu et al. [9] studied the LDS problem on unit disk graphs, and proved that this problem is NP-complete. Furthermore, given an unit disk graph $G = (V, E)$ and an $\epsilon > 0$, they have designed a $(1+\epsilon)$ -factor approximation algorithm to find an LDS in G with running time $n^{O(c^2)}$, where $c = O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$. In this section, we design a $\frac{11}{2}$ -factor approximation algorithm that runs in sub-quadratic time.

For a point $p \in \mathcal{P}$, let $C(p)$ denote the disk centered at the point p . For any two points $p, q \in \mathcal{P}$, if $q \in C(p)$, then we say that q is a neighbor of p (sometimes we say q is covered by p) and vice versa. Since for any Liar's dominating set D , $|(N[p_i] \cup N[p_j]) \cap D| \geq 3$ ($\forall p_i, p_j \in \mathcal{P}$) holds, we assume that $|\mathcal{P}| \geq 3$ and for all the points $p \in \mathcal{P}$, $|N[p]| \geq 2$. For a point $p_i \in \mathcal{P}$, let $p_i(x)$ and $p_i(y)$ denote the x and y coordinates of p_i , respectively. Let $\text{Cov}_{\frac{1}{2}}(C(p_i))$, $\text{Cov}_1(C(p_i))$, $\text{Cov}_{\frac{3}{2}}(C(p_i))$ denote the set of points of $\mathcal{P}$ that are inside the circle centered at p_i and of radius $\frac{1}{2}$, 1 and $\frac{3}{2}$ unit, respectively.

The basic idea of our algorithm is as follows. Initially, we sort the points of $\mathcal{P}$ based on their x -coordinates. Now, consider the leftmost point (say p_i).

We compute the sets $\text{Cov}_{\frac{1}{2}}(C(p_i))$, $\text{Cov}_1(C(p_i))$ and $\text{Cov}_{\frac{3}{2}}(C(p_i))$. Next, we compute the set $Q = \text{Cov}_{\frac{3}{2}}(C(p_i)) \setminus \text{Cov}_1(C(p_i))$. Further, for each point $q_i \in Q$, we compute the set $S(q_i) = \text{Cov}_1(C(q_i)) \cap \text{Cov}_{\frac{1}{2}}(C(p_i))$. Finally, we compute the set $S = \bigcup S(q_i)$. Moreover, our algorithm is divided into two cases.

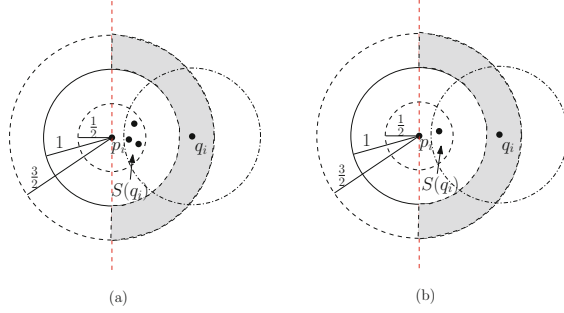


Fig. 1. An illustration of Case 1; (a) $|S(q_i)| \geq 3$, (b) $|S(q_i)| \leq 2$.

Case 1 ($S \neq \emptyset$): For each point $q_i \in Q$ such that $S(q_i) \neq \emptyset$, we further distinguish between the following cases.

1. If $|S(q_i)| \geq 3$: we pick two arbitrary points from the set $S(q_i)$, and include them in the output set D (see Fig. 1(a)).
2. If $|S(q_i)| \leq 2$: in this case, we select one point p_a and a possible second point p_b in the output set D (see Fig. 1(b)).

Once these points are selected, we remove the remaining points from $\text{Cov}_1(C(q_i))$ at this step from the set Q . Notice that the points that lie in $\text{Cov}_1(C(q_i))$ are already 1-dominated. Later, we can pick those points if required. However, observe that we may choose a point p_i from $\text{Cov}_{\frac{1}{2}}(C(p_i))$ while in Case 1.2. That would not constitute a LDS. So we maintain a counter t in Case 1. This counter keeps track of how many points we are picking from the set $S(q_i)$ in total, for each point $q_i \in Q$. If t is at least 2, we simply add p_i to the output set and do not enter into Case 2. Otherwise, we proceed to Case 2.

Case 2 ($S = \emptyset$ or $t < 2$): here, we further distinguish between the following cases.

1. If $|\text{Cov}_{\frac{1}{2}}(C(p_i))| \geq 3$: then we choose 2 points arbitrarily in the output set D (see Fig. 2(a)).
2. If $|\text{Cov}_{\frac{1}{2}}(C(p_i))| = 2$: let $p_i, p_x \in \text{Cov}_{\frac{1}{2}}(C(p_i))$ be these points. We include both of them in the output set D . This settles the first condition of LDS for them. However, in order to fulfill the second condition of LDS for p_i and p_x , we must include at least one extra point here. First, we check the cardinality of $X = (\text{Cov}_1(C(p_i)) \cap \text{Cov}_1(C(p_x))) \setminus \{p_i, p_x\}$. If $|X| \neq \emptyset$, then we pick an

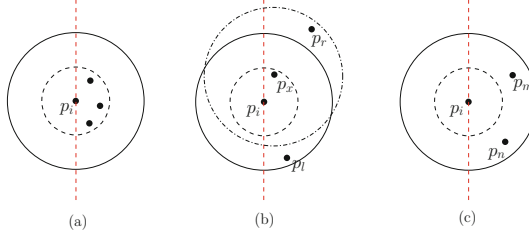


Fig. 2. An illustration of Case 2; (a) $|\text{Cov}_{\frac{1}{2}}(C(p_i))| \geq 3$, (b) $|\text{Cov}_{\frac{1}{2}}(C(p_i))| = 2$, (c) $|\text{Cov}_{\frac{1}{2}}(C(p_i))| = 1$.

arbitrary point p_m from X , and include p_m in D . Otherwise, we include two points $p_l \in \text{Cov}_1(C(p_i))$ and $p_r \in \text{Cov}_1(C(p_x))$, and include them in D (see Fig. 2(b)). Note that, in this case, we know that p_l and p_r exist due to the input constraint of an LDS problem.

3. If $|\text{Cov}_{\frac{1}{2}}(C(p_i))| = 1$: then we check $\text{Cov}_1(C(p_i))$ and include two points p_m and p_n arbitrarily from $\text{Cov}_1(C(p_i)) \setminus \text{Cov}_{\frac{1}{2}}(C(p_i))$ (see Fig. 2(c)).

This fulfills the criteria of LDS of points in $\text{Cov}_{\frac{1}{2}}(C(p_i))$. Then we delete the remaining points of $\text{Cov}_{\frac{1}{2}}(C(p_i))$ from $\mathcal{P}$. Next we select the left-most point from the remaining and repeat the same procedure until $\mathcal{P}$ is empty. The pseudo-code of the algorithm is given in the full version of the paper (see [1]).

Lemma 1. $[\star]^1$ *The set D obtained from our algorithm, is a LDS of the unit disk graph defined on the points of $\mathcal{P}$.*

Lemma 2. $[\star]$ *Our algorithm outputs a LDS $D \subseteq \mathcal{P}$ of the unit disk graph defined on the points of $\mathcal{P}$ with approximation ratio $\frac{11}{2}$.*

The algorithm runs in polynomial time (to be precise in sub-quadratic time). Thus, from Lemmas 1 and 2 we conclude the following theorem.

Theorem 1. *The algorithm computes a LDS of the unit disk graph defined on the points of $\mathcal{P}$ in sub-quadratic time with approximation factor $\frac{11}{2}$.*

2.2 PTAS for k -DS on Unit Disk Graphs

In this section we give a PTAS for the k -tuple dominating set on unit disk graphs with a similar approach used by Nieberg and Hurink [13]. It might be possible to design a PTAS by using local search or shifting strategy for the k -tuple dominating set problem on unit disk graphs. However, the time complexity of these algorithms would be high. Thus we use the approach of Nieberg and Hurink [13], that guarantees a better running time.

¹ Proof of results labeled with $[\star]$ have been deferred to the full version [1] due to space constraint.

Let $G = (V, E)$ be an unit disk graph in the plane. For a vertex $v \in V$, let $N^r(v)$ and $N^r[v] = N^r(v) \cup \{v\}$ be the r -th neighborhood and r -th closed neighborhood of v , respectively. For any two vertices $u, v \in V$, let $\delta(u, v)$ be the distance between u and v in G , that is the number of edges of a shortest path between u and v in G . Let $D_k(V)$ be the minimum k -tuple dominating set of G . For a subset $W \subseteq V$, let $D_k(W)$ be the minimum k -tuple dominating set of the induced subgraph on W . We prove the following theorem.

Theorem 2. *There exists a PTAS for the k -tuple dominating set problem on unit disk graphs.*

Proof. The 2-separated collection of subsets is defined as follows: Given a graph $G = (V, E)$, let $S = \{S_1, \dots, S_m\}$ be a collection of subsets of vertices $S_i \subset V$, for $i = 1, \dots, m$, such that for any two vertices $u \in S_i$ and $v \in S_j$ with $i \neq j$, $\delta(u, v) > 2$. In the following lemma we prove that the sum of the cardinalities of the minimum k -tuple dominating sets $D_k(S_i)$ for the subsets $S_i \in S$ of a 2-separated collection is a lower bound on the cardinality of $D_k(V)$.

Lemma 3. $[\star]$ *Given a graph $G = (V, E)$, let $S = \{S_1, \dots, S_m\}$ be a 2-separated collection of subsets of V then, $|D_k(V)| \geq \sum_{i=1}^m |D_k(S_i)|$.*

From Lemma 3, we get the lower bound of the minimum k -tuple dominating set of G . If we can enlarge each of the subset S_i to a subset T_i such that the k -tuple dominating set of S_i (that is $D_k(S_i)$) is locally bounded to the k -tuple dominating set of T_i (that is $D_k(T_i)$), then by taking the union of them we get an approximation of the k -tuple dominating set of G . For each subset S_i , let there is a subset T_i (where $S_i \subset T_i$), and let there exists a bound $(1 + \epsilon)$ ($0 < \epsilon < 1$) such that $|D_k(T_i)| \leq (1 + \epsilon) \cdot |D_k(S_i)|$. Then, if we take the union of the k -tuple dominating sets of all T_i , this is a $(1 + \epsilon)$ -approximation of the k -tuple dominating sets of the union of subsets S_i (for $i = 1, \dots, m$). Now, we describe the algorithm. Let $V_0 = V$. Consider an arbitrary vertex $v \in V_0$, and begin computing the k -DS of $N^r[v]$, until $D_k(N^{r+2}[v]) > \rho \cdot D_k(N^r[v])$ (for a constant ρ). We iteratively process the remaining graph induced by $V_{i+1} = V_i \setminus N^{\hat{r}_i+2}[v_i]$ (where $\hat{r}_i$ is the first point at i th iteration when the condition is violated).

Lemma 4. *Let $\{N_1, \dots, N_\ell\}$ be the set of neighborhoods created by the above algorithm (for $\ell < n$). The union $\bigcup_{i=1}^\ell (D_k(N_i))$ forms a k -tuple dominating set of G .*

Proof. Consider the set $V_{i+1} = V_i \setminus N_i$, and we know $N_i \subset V_i$. Thus, $V_{i+1} = V_i \cup N_i$. The algorithm stops while $V_{\ell+1} = \emptyset$, which means $V_\ell = N_\ell$. Besides, $\bigcup_{i=1}^\ell (N_i) = V$. Thus, if we compute the k -tuple dominating set $D_k(N_i)$ of each N_i , their union clearly is the k -tuple dominating set of the entire graph. $\square$

These subsets $N^{\hat{r}_i}[v_i]$, for $i = 1, \dots, \ell$, created by the algorithm form a 2-separated collection $\{N^{\hat{r}_1}[v_1], \dots, N^{\hat{r}_\ell}[v_\ell]\}$ in G . Consider any two neighborhoods N_i, N_{i+1} . We have computed N_{i+1} on graph induced by $V \setminus V_i$. So, for any two vertices $u \in N_i$ and $v \in N_{i+1}$, the distance is greater than 2. Thus we have the following corollary.

Corollary 1. *The algorithm returns a k -tuple dominating set $\bigcup_{i=1}^{\ell} (D_k(N_i))$ of cardinality no more than ρ the size of a dominating set $D_k(V)$, where $\rho = (1 + \epsilon)$.*

It needs to be shown that this algorithm has a polynomial running time. The number of iterations ℓ is clearly bounded by $|V| = n$. It is important to show that for each iteration we can compute the minimum k -tuple dominating set $D_k(N^r[v])$ in polynomial time for r being constant or polynomially bounded. Consider the r -th neighborhood of a vertex v , $N^r[v]$. Let I^r be the maximal independent set of the graph induced by $N^r[v]$. From [13], we have $|I^r| \leq (2r + 1)^2 = O(r^2)$. The cardinality of a minimum dominating set in $N^r[v]$ is bounded from above by the cardinality of a maximal independent set in $N^r[v]$. Hence, $|D(N^r[v])| \leq (2r + 1)^2 = O(r^2)$. Now, we prove the following lemma.

Lemma 5. $|D_k(N^r[v])| \leq O(k^2 \cdot r^2)$.

Proof. Let I_1^r be the first maximal independent set of $N^r[v]$. We know $|D(N^r[v])| \leq |I_1^r| \leq (2r + 1)^2$. Now, we take the next maximal independent set I_2^r from $N^r[v] \setminus I_1^r$, and take the union of them $(I_1^r \cup I_2^r)$. Notice that every vertex $v \in (N^r[v] \setminus (I_1^r \cup I_2^r))$ has 2 neighbors in $(I_1^r \cup I_2^r)$, so they can be 2-tuple dominated by choosing vertices from $(I_1^r \cup I_2^r)$. Also, every vertex $v \in I_2^r$ can be 2-tuple dominated by choosing vertices from $(I_1^r \cup I_2^r)$, since v itself can be one and the other one can be picked from I_1^r . Additionally, for each vertex $u \in I_1^r$, we take a vertex z from the neighborhood of u in $(N^r[v] \setminus (I_1^r \cup I_2^r))$. Let $W(I_1^r)$ be the union of these vertices. Now, every vertex $v \in N^r[v]$ can be 2-tuple dominated by choosing vertices from $(I_1^r \cup I_2^r \cup W(I_1^r))$. So, $|D_2(N^r[v])| \leq |(I_1^r \cup I_2^r \cup W(I_1^r))|$. $|I_1^r \cup I_2^r \cup W(I_1^r)| \leq 3 \cdot (2r + 1)^2$. Hence, $|D_2(N^r[v])| \leq 3 \cdot (2r + 1)^2$. We continue this process k times.

After k steps, we get the union of the maximal independent sets $A = \{I_1^r \cup \dots \cup I_k^r\}$. Additionally, we get the unions of $B = \{W(I_1^r) \cup W(I_1^r \cup I_2^r) \cup \dots \cup W(I_1^r \cup \dots \cup I_{k-1}^r)\}$. Notice that every vertex $v \in N^r[v]$ can be k -tuple dominated by choosing vertices from $(A \cup B)$. The cardinality of $(A \cup B)$ is at most $(2r + 1)^2 \cdot (1 + 3 + \dots + (2k - 1))$, which is $(2r + 1)^2 \cdot k^2$. We also know $|D_k(N^r[v])|$ is upper bounded by $(A \cup B)$. Thus, $|D_k(N^r[v])| \leq (2r + 1)^2 \cdot k^2 \leq O(k^2 \cdot r^2)$.

Nieberg and Hurink [13] showed that for a unit disk graph, there exists a bound on $\hat{r}_1$ (the first value of r that violates the property $D(N^{r+2}[v]) > \rho \cdot D(N^r[v])$). This bound depends on the approximation ρ not on the size of the of the unit disk graph $G = (V, E)$ given as input. Precisely, they have proved that there exists a constant $c = c(\rho)$ such that $\hat{r}_1 \leq c$, that is, the largest neighborhood to be considered during the iteration of the algorithm is bounded by a constant. Thereby, putting everything together, we conclude the proof. $\square$

3 Hardness Results

3.1 Streaming Lower Bound for LDS

In this section, we consider the streaming model: the edges arrive one-by-one in some order, and at each time-stamp we need to decide if we either *store* the edge

or *forget* about it. We now show that any streaming algorithm that solves the LDS problem must essentially store all the edges.

Theorem 3. *Any randomized² streaming algorithm for LDS problem on n -vertex graphs requires $\Omega(n^2)$ space.*

Proof. We will reduce from the INDEX problem in communication complexity:

Index Problem

Input: Alice has a string $X \in \{0, 1\}^N$ given by $x_1 x_2 \dots x_N$. Bob has an index $\iota \in [N]$.

Question: Bob wants to find x_ι , i.e., the ι^{th} bit of X .

It is well-known that there is a lower bound of $\Omega(N)$ bits in the one-way randomized communication model for Bob to compute x_i [11]. We assume an instance of the INDEX problem where N is a perfect square, and let $r = \sqrt{N}$. Fix any bijection from $[N] \rightarrow [r] \times [r]$. Consequently we can interpret the bit string as an adjacency matrix for a bipartite graph with r vertices on each side. Let the two sides of the bipartition be $V = \{v_1, v_2, \dots, v_r\}$ and $W = \{w_1, w_2, \dots, w_r\}$.

From the instance of INDEX, we construct an instance G_X of the LDS. Assume that Alice has an algorithm that solves the k -tuple dominating set problem using $f(r)$ bits. First, we insert the edges corresponding to the edge interpretation of X between nodes v_i and w_j : for each $i, j \in [r]$, Alice adds the edge (v_i, w_j) if the corresponding entry in X is 1. Alice then sends the memory contents of her algorithm to Bob, using $f(r)$ bits.

Bob has the index $\iota \in [N]$, which he interprets as (I, J) under the same bijection $\phi : [N] \rightarrow [r] \times [r]$. He receives the memory contents of the algorithm, and proceeds to do the following:

- Add two vertices a and b , and an edge $a - b$
- Add an edge from each vertex of $V \setminus v_I$ to a
- Add an edge from each vertex of $W \setminus w_J$ to b
- Add five vertices $\{u, y, u', y', z\}$ and edges $u - u', y - y', u - z$ and $y - z$.
- Add an edge from each vertex of $V \cup W \cup \{a, b\}$ to each vertex from $\{u, y\}$

Let D be a minimum LDS of G_X . Note that D has to be a double dominating set of G_X . Since u' has only 2 neighbors in G_X , it follows that $\{u, u'\} \subseteq D$. Similarly $\{y, y'\} \subseteq D$. Note that z also has only two neighbors in G_X . Hence, we have that $N[z] \cup N[b] = \{u, y, a, b\}$. Since we must have $|(N[z] \cup N[b]) \cap D| \geq 3$, it follows that at least one of a or b must belong in D . Since $N[b] \subseteq N[a]$, without loss of generality we can assume that $a \in D$. Therefore, so far we have concluded that $\{u, u', y, y', a\} \subseteq D$.

The next two lemmas show that finding the minimum value of a LDS of G_X allows us to solve the corresponding instance X of INDEX.

² By randomized algorithm we mean that the algorithm should succeed with probability $\geq \frac{2}{3}$.

Lemma 6. $x_i = 1$ implies that the minimum size of a LDS of G_X is 6.

Proof. Suppose that $x_i = 1$, i.e., $v_I - w_J$ is an edge in G_X . We now claim that $D := \{u, u', y, y', a\} \cup v_I$ is a LDS of G_X .

First we check that D is indeed a double dominating set of G_X

- For each vertex in $\lambda \in G_X \setminus \{u, u', y, y', z\}$ we have $(N[\lambda] \cap D) \supseteq \{u, y\}$
- $(N[u] \cap D) \supseteq \{u, u'\}$
- $(N[y] \cap D) \supseteq \{y, y'\}$
- $(N[z] \cap D) = \{u, y\}$
- $(N[u'] \cap D) = \{u, u'\}$
- $(N[y'] \cap D) = \{y, y'\}$

We now check the second condition. Let $T = G_X \setminus \{u, u', y, y', z\}$, and $T' = G_X \setminus T$

- For each $\lambda \in T \setminus \{v_I, w_J\}$ and each $\delta \in T'$ we have $(N[\lambda] \cup N[\delta]) \cap D \supseteq \{a, u, y\}$
- For each $\delta \in T'$ we have $(N[v_I] \cup N[\delta]) \cap D \supseteq \{v_I, u, y\}$
- For each $\delta \in T'$ we have $(N[w_J] \cup N[\delta]) \cap D \supseteq \{v_I, u, y\}$
- Now we consider pairs where both vertices are from T' . By symmetry, we only have to consider following choices
 - $(N[u'] \cup N[u]) \cap D = \{u, u', y\}$
 - $(N[u'] \cup N[z]) \cap D = \{u, u', y\}$
 - $(N[u'] \cup N[y]) \cap D = \{u, u', y, y'\}$
 - $(N[u'] \cup N[y']) \cap D = \{u, u', y, y'\}$
- Now we consider pairs where both vertices are from T . By symmetry, we only have to consider following choices
 - For each $\lambda \in V \setminus v_I \cup W \setminus w_J$ we have $(N[\lambda] \cup N[b]) \cap D \supseteq \{u, y, a\}$ and $(N[\lambda] \cup N[a]) \cap D \supseteq \{u, y, a\}$
 - $(N[v_I] \cup N[b]) \cap D \supseteq \{u, y, a\}$
 - $(N[v_I] \cup N[a]) \cap D \supseteq \{u, y, a\}$
 - $(N[w_J] \cup N[b]) \cap D \supseteq \{u, y, a\}$
 - $(N[w_J] \cup N[a]) \cap D \supseteq \{u, y, a\}$
 - $(N[w_J] \cup N[v_I]) \cap D \supseteq \{v_I, y, a\}$
 - For each $\gamma \in V \setminus v_I$ we have $(N[w_J] \cup N[\gamma]) \cap D \supseteq \{v_I, y, a, u\}$
 - For each $\gamma \in W \setminus w_J$ we have $(N[v_I] \cup N[\gamma]) \cap D \supseteq \{v_I, y, a, u\}$
 - For each $\gamma \in V \setminus v_I$ and $\gamma' \in W \setminus w_J$ we have $(N[\gamma] \cup N[\gamma']) \cap D \supseteq \{y, a, u\}$

Hence, it follows that D is indeed a LDS of G_X of size 6.

Lemma 7. $x_i = 0$ implies that the minimum size of a LDS of G_X is ≥ 7 .

Proof. Now suppose that $x_i = 0$, i.e., v_I and w_J do not have an edge between them in G_X . Let D' be a minimum LDS of G_X . We have already seen above that $\{u, u', y, y', a\} \subseteq D$.

Consider the pair (v_I, z) . Currently, we have that $(N[v_I] \cup N[z]) \cap \{u, u', y, y', a\} = \{u, y\}$. Hence, D' must contain a vertex, say $\mu \in N[v_I] \setminus \{u, y\}$. Consider the pair (w_J, z) . Currently, we have that $(N[w_J] \cup N[z]) \cap$

$\{u, u', y, y', a\} = \{u, y\}$. Hence, D' must contain a vertex, say $\mu' \in N[w_J] \setminus \{u, y\}$. Since v_I and w_J do not form an edge, we have that $\mu \neq \mu'$. Hence, $|D'| \geq 5 + 2 = 7$.

Thus, by checking whether the value of a minimum LDS on the instance G_X is 6 or 7, Bob can determine the index x_i . The total communication between Alice and Bob was $O(f(r))$ bits, and hence we can solve the INDEX problem in $f(r)$ bits. Recall that the lower bound for the INDEX problem is $\Omega(N) = \Omega(r^2)$. Note that $|G_X| = n = 2r + 5 = O(r)$, and hence $\Omega(r^2) = \Omega(n^2)$. $\square$

Corollary 2. *Let $\epsilon > 0$ be a constant. Any (randomized) streaming algorithm that achieves a $(\frac{7}{6} - \epsilon)$ -approximation for a LDS requires $\Omega(n^2)$ space.*

Proof. Theorem 3 shows that distinguishing between whether the minimum value of the LDS is 6 or 7 requires $\Omega(n^2)$ bits. The claim follows since $6 \cdot (\frac{7}{6} - \epsilon) < 7$. $\square$

3.2 Streaming Lower Bounds for k -DS

Theorem 4. *For any $k = O(1)$, any randomized (See footnote 2) streaming algorithm for the k -tuple dominating set problem on n -vertex graphs requires $\Omega(n^2)$ space.*

Proof. We reduce from the INDEX problem in communication complexity. We assume an instance of the INDEX problem where N is a perfect square, and let $r = \sqrt{N}$. Fix any bijection from $[N] \rightarrow [r] \times [r]$. Consequently we can interpret the bit string as an adjacency matrix for a bipartite graph with r vertices on each side. Let the two sides of the bipartition be $V = \{v_1, v_2, \dots, v_r\}$ and $W = \{w_1, w_2, \dots, w_r\}$. From the instance of INDEX, we construct an instance G_X of the k -tuple dominating set. Assume that Alice has an algorithm that solves the k -tuple dominating set problem using $f(r)$ bits. First, we insert the edges corresponding to the edge interpretation of X between nodes v_i and w_j : for each $i, j \in [k]$, Alice adds the edge (v_i, w_j) if the corresponding entry in X is 1. Alice then sends the memory contents of her algorithm to Bob, using $f(r)$ bits.

Bob has the index $\iota \in [N]$, which he interprets as (I, J) under the same bijection $\phi : [N] \rightarrow [r] \times [r]$. He receives the memory contents of the algorithm, and proceeds to do the following:

- Add $(k+1)$ vertices $A = \{a_1, a_2, \dots, a_k\}$ and b .
- Add edges $\{a_i - b : 1 \leq i \leq k\}$.
- Add an edge from each vertex of $V \setminus (v_I)$ to each vertex of A .
- Add an edge from each vertex of $W \setminus (w_J)$ to each vertex of A .
- Add an edge v_I to each vertex of $A \setminus a_k$.
- Add an edge from w_J to each vertex of $A \setminus a_k$.

The next lemma shows that finding the minimum value of a k -tuple dominating set of G_X allows us to solve the corresponding instance X of INDEX.

Lemma 8. $[\star]$ *The minimum size of a k -tuple dominating set of G_X is $k + 1$ if and only if $x_i = 1$.*

Thus, by checking whether the value of minimum k -tuple dominating set on the instance G_X is $k + 1$ or $k + 2$, Bob can determine the index x_i . The total communication between Alice and Bob was $O(f(r))$ bits, and hence we can solve the INDEX problem in $f(r)$ bits. Recall that the lower bound for the INDEX problem is $\Omega(N) = \Omega(r^2)$. Note that $|G_X| = n = 2r + k + 1 = O(r)$ since $k = O(1)$, and hence $\Omega(r^2) = \Omega(n^2)$.

Corollary 3. *Let $1 > \epsilon > 0$ be any constant. Any (randomized) streaming algorithm that approximates a k -tuple dominating set within a relative error of ϵ requires $\Omega(n^2)$ space.*

Proof. Choose $\epsilon = \frac{1}{k}$. Theorem 8 shows that the relative error is at most $\frac{1}{k+2}$, which is less than ϵ . Hence finding an approximation within ϵ relative error amounts to finding the exact value of the k -tuple dominating set. Hence, the claim follows from the lower bound of $\Omega(n^2)$ of Theorem 8.

3.3 W-Hardness Results for LDS

The LDS problem is NP-complete on general graphs [17], and NP-complete on bipartite graphs, split graphs (see, e.g., [14, 16]). Bishnu et al. [2] showed that LDS problem on planar graphs admits a linear kernel and W[2]-hard on general graphs. We prove that LDS problem is W[2]-hard on bipartite graphs.

Theorem 5. *Liar's dominating set on bipartite graphs is W[2]-hard.*

Proof. We prove this by giving a parameterized reduction from the dominating set problem in general undirected graphs. Let $(G = (V, E), k)$ be an instance of the dominating set, where k denotes the size of the dominating set. We construct a bipartite graph $G' = (V', E')$ from $G = (V, E)$. First, we create two copies of V , namely $V_1 = \{u_1 | u \in V\}$ and $V_2 = \{u_2 | u \in V\}$. Next, we introduce two extra vertices z_1, z_2 in V_1 , and two extra vertices z'_1, z'_2 in V_2 . Furthermore, we introduce two special vertices $s_{z'_1}, s_{z'_2}$ in V_1 and two special vertices s_{z_1}, s_{z_2} in V_2 . The entire vertex set V' is $V_1 \cup V_2$, where $V_1 = \{u_1 | u \in V\} \cup \{z_1, z_2, s_{z'_1}, s_{z'_2}\}$ and $V_2 = \{u_2 | u \in V\} \cup \{z'_1, z'_2, s_{z_1}, s_{z_2}\}$. Now, if there is an edge $(u, v) \in E$, then we draw the edges (u_1, v_2) and (v_1, u_2) . We draw the edges of the form (u_1, u_2) in G' for every vertex $u \in V$. Then, we add edges from every vertex in $V_1 \setminus \{z_1, z_2, s_{z'_1}, s_{z'_2}\}$ to z'_1, z'_2 , and the edges from every vertex in $V_1 \setminus \{z'_1, z'_2, s_{z_1}, s_{z_2}\}$ to z_1, z_2 . Finally we add the edges $(z_1, z'_1), (z_2, z'_2)$ and $(z_1, s_{z_1}), (z_2, s_{z_2}), (z'_1, s_{z'_1}), (z'_2, s_{z'_2})$. This completes the construction (see Fig. 3).

We show that G has a dominating set of size k if and only if G' has a LDS of size $k + 8$. Let D denote the dominating set of the given graph G . We claim that $D' = \{u_1 | u \in D\} \cup \{z_1, z_2, s_{z'_1}, s_{z'_2}\} \cup \{z'_1, z'_2, s_{z_1}, s_{z_2}\}$ is a LDS of G' . Note that for any vertex $v \in V'$, $|N_{G'}[v] \cap D'| \geq 2$, since $\{z_1, z_2, z'_1, z'_2\}$ is in D' . This fulfills the first condition of the LDS. Now, for every pair of vertices, $u, v \in V'$, we show that $|(N_{G'}[u] \cup N_{G'}[v]) \cap D'| \geq 3$. If $u, v \in V_1$, we know $|(N_{G'}[u] \cup N_{G'}[v]) \cap D'| \geq 2$ due to z'_1, z'_2 . Now, in the dominating set at least one additional vertex dominates

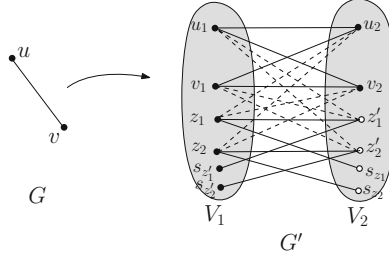


Fig. 3. Construction of G' from G (Illustration of Theorem 5).

them. Thus, $|(N_{G'}[u] \cup N_{G'}[v]) \cap D'| \geq 3$. Similarly, when $u, v \in V_2$ or $u \in V_1$ and $v \in V_2$. This fulfills the second condition of the LDS.

Conversely, let D' be a LDS in G' . Note that, $\{z_1, z_2, z'_1, z'_2\}$ are always part of D' , since z_1, z_2 are the only neighbors of s_{z_1}, s_{z_2} and z'_1, z'_2 are the only neighbors of $s_{z'_1}, s_{z'_2}$. These special vertices are taken in the construction to enforce $\{z_1, z_2, z'_1, z'_2\}$ to be in D' . Now we know, for any pair of vertices p, q , $|(N_{G'}[p] \cup N_{G'}[q]) \cap D'| \geq 3$. This implies p, q is dominated by at least one vertex or one of them is picked, except $\{z_1, z_2, z'_1, z'_2\}$. Otherwise, $|(N_{G'}[p] \cup N_{G'}[q]) \cap D'| < 3$. This violates the second condition of LDS. Now, when p, q are both part of the same edge in G (say $u_2, v_2 \in V_2$; see Fig. 3), we need at least one vertex from $\{u_1, v_1, u_2, v_2\}$ in D' . This means that for every vertex $v \in V$, $|N_G[v] \cap D| \geq 1$. Thus, D is a dominating set of G where the cardinality of D is at most k . $\square$

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